

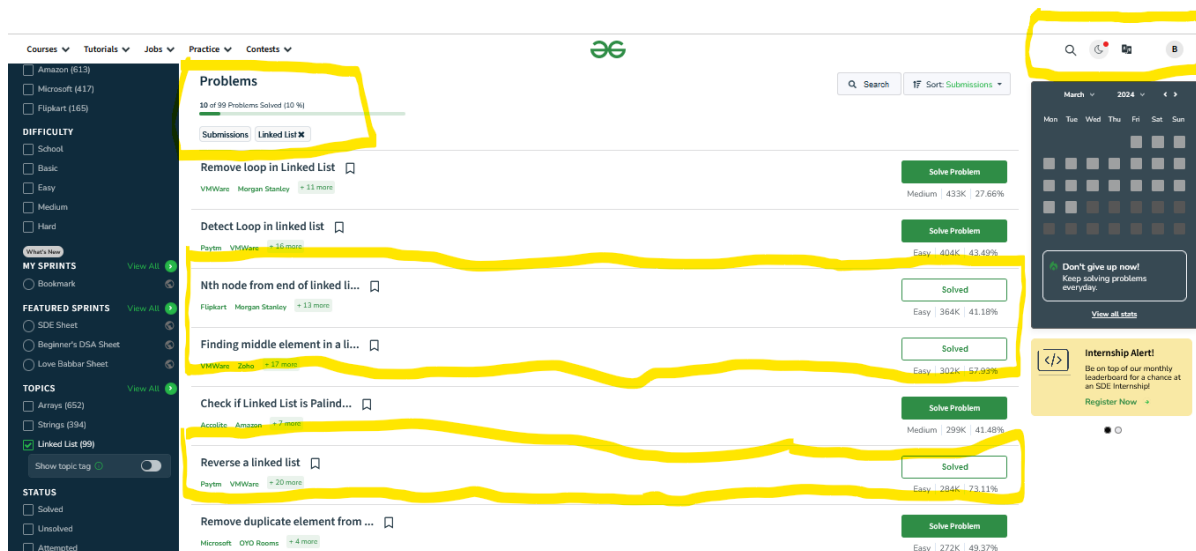
DSA Assignment 1

Create your account on “geeksforgeeks.org” and solve below programming questions from the two sets as:

- **Set 1:** All questions are compulsory for all students. Section A carries 5 marks and Section B carries 10 marks. Answers for Section B must be written on a separate sheet and uploaded to the LMS.
- **Set 2:** Check your name and solve the assigned problem only. (5 Marks)

Instructions for Submission: Submit the assignment in hardcopy in two pages as:-

Page 1 should be cover page and page 2 should highlight your profile name, the number of questions you have solved and the questions from Set 2 which you have solved (Refer example screenshot below).



Important Note: The assignment has been designed with serious efforts and with the sole purpose of motivating you to solve competitive programming while learning the concepts in the classroom which will enable you to score better in your actual placements.

You are required to put your sincere efforts solving them, however if the tasks couldn't be solved in given time, appropriate marks will be awarded. For the cases of copying, unfair means or taking AI help, please note that it will be severely punished with negative marking.

The assignments will be evaluated in the classroom one by one, for your turn keep the hard copy and your laptop ready for the evaluation.

Set 1

Section A

SNO	Tasks
1	Remove loop in Linked List
2	Detect Loop in linked list
3	Delete a Node in Single Linked...
4	Linked List Insertion
5	Count nodes of linked list
6	Check If Circular Linked List
7	Find length of Loop
8	Print Linked List elements
9	Reverse a Doubly Linked List
10	Linked List Length Even or Odd...

SET-1 SECTION B

- Q1.** Differentiate between static and dynamic memory allocation with one example each.
- Q2.** What is an Abstract Data Type (ADT)? Why is it important in software design?
- Q3.** Explain the concept of a function pointer as a member of a structure in C with a brief syntax example.
- Q4.** Explain the row-major and column-major representations of a 2D array in memory. Derive the address formula for element $A[i][j]$ stored in row-major order. Assume base address B, element size S, and dimensions $m \times n$.
- Q5.** Compare iterative and recursive implementations of binary search. Write pseudocode for both, analyze their time and space complexities, and state when one is preferred over the other.
- Q6.** Describe with neat diagrams all operations (insertion at beginning, end, and middle; deletion) on a doubly linked list. Show pointer manipulations clearly.
- Q7.** What is a generalized linked list? Explain how it differs from a standard linked list and illustrate its use in representing a polynomial with multiple variables (e.g., $3x^2y + 2xy^2 - 5$).
- Q8.** Discuss the following with respect to Analysis of Algorithms:
- a)** Amortized Analysis: Define and explain aggregate, accounting, and potential methods with a suitable example for each.
 - b)** Asymptotic Notations: Define O , Ω , and Θ notations. Illustrate with graphs and prove that $f(n) = 3n^2 + 5n + 2$ is $O(n^2)$.
- Q9.** Write and explain complete C algorithms for the following array operations on an unsorted array of integers:
- a)** Insertion of an element at a specified position.
 - b)** Deletion of a specified element (first occurrence).
 - c)** Linear and Binary Search — write code for both and compare their best, worst, and average case complexities in a table.
- Q10.** Explain the Skip List data structure in detail:
- a)** Draw the structure of a skip list containing keys {2, 5, 8, 12, 17, 21} with 3 levels. Explain how the probabilistic promotion of nodes works.
 - b)** Write the search algorithm for a skip list and analyse its expected time complexity. Compare skip lists with singly linked lists and balanced BSTs in terms of time complexity for search, insertion, and deletion. Present your comparison in a table.

Set-2

SNO	Tasks	Student Name (batch 9 &10)	
1	Nth node from end of linked li...	Harshit Kumar Agarwal	
2	Finding middle element in a li...	Prateek Prateek	
3	Reverse a linked list	Parikshit Bansal	
4	Remove duplicate element from ...	Murchana Gogoi	
5	Rotate a Linked List	Pratik Agarwal	
6	Delete without head pointer	Dia Pal	
7	Add 1 to a number represented ...	Aadvit More	
8	Given a linked list of 0s, 1s ...	Tarush Singh	
9	Merge two sorted linked lists	Satyam Rastogi	
10	Flattening a Linked List	Ayan Jain	
11	Delete nodes having greater va...	Rachit Madhar	
12	Intersection of two sorted Lin...	Bhavin S	
13	Insert in a Sorted List	Saksham Saxena	
14	Identical Linked Lists	Rudram Sharma	
15	Sorted insert for circular lin...	Pratibha Singh Ranawat	
16	Count Pairs whose sum is equal...	Bhavini Meena	
17	Insert in Middle of Linked Lis...	Arunav Duggal	
18	Delete Middle of Linked List	Himansh Pandey	
19	Node at a given index in linke...	Harshita Pandey	
20	Delete node in Doubly Linked L...	Prikshit	
21	Merge K sorted linked lists	Piyush Mani Tripathi	
22	Subtraction in Linked List	Sanvi Kaur Bawa	
23	Delete Alternate Nodes	Parth Singh	
24	Merge 2 sorted linked list in ...	Yashvi Singh	
25	Intersection of Two Linked Lis...	Anik Suhag	
26	Reorder List	Ankit Kumar Yadav	
27	Occurence of an integer in a L...	Kavya Agrawal	
28	Union of Two Linked Lists	Anshuman Anshuman	
29	Linked List that is Sorted Alt...	Somiya Chhawri	
30	Reverse alternate nodes in Lin...	Deepanshi Shukla	
31	Swap Kth nodes from ends	Arjit Pathak	
32	Remove every k'th node	Dushyant	
33	Split a Circular Linked	Ayush Pandey	

	List 1...		
34	Delete N nodes after M nodes o...	Vinayak Dubey	
35	Find n/k th node in Linked lis...	Ayushi Mishra	
36	Multiply two linked lists	Manas Mishra	
37	Rearrange a linked list	Hiya Nihar Dave	
38	Find the first node of loop in...	Shresth Deb Roy	
39	Find the Sum of Last N nodes o...	Divyansh Bhatia	
40	Insertion Sort for Singly Link...	Grikzhana Daulagupu	
41	Move Last Element to Front of ...	Venya Mehrotra	
42	Modular Node	Kashvi Goyal	
43	Insert in Sorted way in a Sort...	Divyanshi Bhatt	
44	Deletion and Reverse in Linked...	Tejas Khari	
45	Prime List	Sachin Kumar	
46	Compare two linked lists	Sidak Singh Saluja	
47	Modify Linked List-I	Nidhish Rawat	
48	Length of longest palindrome i...	Dhruv Jha	
49	linked list of strings forms a...	Bhavik Benner	
50	Reverse both parts	Sumit Ranjan Jha	
51	Move all zeros to the front of...	Nishika Yadav	
52	Partition a Linked List around...	Radhika Mehrotra	
53	Add two numbers represented by...	Shivansh Upadhyay	
54	Remove all occurrences of dupli...	Aadarsh Tyagi	
55	Split Singly Linked List Alter...	Harshita Mathur	